
PHYSICS

9702/51

Paper 5 Planning, Analysis and Evaluation

October/November 2017

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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Question	Answer	Marks
1	Defining the problem	
	d is the independent variable and R is the dependent variable or vary d and measure R	1
	keep intensity/power of light source <u>constant</u>	1
	Methods of data collection	
	labelled diagram showing a light source fixed above container of water with the labelled LDR positioned in the beaker	1
	correct circuit diagram to measure R , e.g. V and I methods or ohmmeter	1
	method to determine R , e.g. $R = \frac{\text{p.d. across LDR}}{\text{current}}$ or read off ohmmeter	1
	method to determine d , e.g. use a ruler or drawn labelled vertical ruler adjacent to container <u>with d indicated</u>	1
	Method of analysis	
	plots a graph of R against d^2	1
	relationship valid if a straight line produced <u>passing through the origin</u>	1
	$K = \frac{4\pi}{\text{gradient}}$	1

Question	Answer	Marks
1	Additional detail including safety considerations	Max. 6
D1	dark glasses to prevent damage to eyes due to light source or do not look directly at light source or do not touch <u>hot</u> lamp/use gloves to position <u>hot</u> lamp/ <u>heat-proof</u> gloves to position lamp	
D2	dark room or shielding LDR (so as to avoid light from other sources)	
D3	use high intensity lamp or collimated beam or laser	
D4	<u>method</u> described to check that current in light source is constant, e.g. use an ammeter and variable resistor / variable power supply	
D5	keep position of light source constant or distance between light source and LDR constant	
D6	light source is placed close to water surface to increase intensity/reduce reflections or light source is placed further away to make it more directional	
D7	use tall container <u>to give a wide range of d or R</u> or <u>to reduce uncertainties</u> or use a wide container <u>to reduce reflections</u>	
D8	method to position ruler vertically to measure d described e.g. use a set square/spirit level	
D9	use of horizontal fiducial mark from ruler to meniscus or middle of LDR, e.g. pin or d = reading on rule at surface – reading at top of LDR	
D10	ensure that the electrical connections/wire to the LDR are waterproof	

Question	Answer	Marks							
2(a)	$\text{gradient} = -\frac{2mg}{s}$ $y\text{-intercept} = mg$	1							
2(b)	<table><tr><td>$(T_1 - T_2)/N$</td></tr><tr><td>5.5 ± 0.2</td></tr><tr><td>4.6 ± 0.2</td></tr><tr><td>3.6 ± 0.2</td></tr><tr><td>2.8 ± 0.2</td></tr><tr><td>1.9 ± 0.2</td></tr><tr><td>1.3 ± 0.2</td></tr></table> First mark for column heading and values of $(T_1 - T_2)/N$. Second mark for all uncertainties = ± 0.2.	$(T_1 - T_2)/N$	5.5 ± 0.2	4.6 ± 0.2	3.6 ± 0.2	2.8 ± 0.2	1.9 ± 0.2	1.3 ± 0.2	2
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2(c)(i)	Six points plotted correctly. Must be within half a small square. Diameter of points must be less than half a small square	1							
	Error bars in P plotted correctly. All error bars to be plotted. Length of bar must be accurate to less than half a small square and symmetrical.	1							

Question	Answer	Marks
2(c)(ii)	Line of best fit drawn. Must not be drawn from top point to bottom point. If points are plotted correctly then upper end of line should pass between (0.125, 5.0) and (0.140, 5.0) and lower end of line should pass between (0.360, 1.5) and (0.380, 1.5).	1
	Worst acceptable line drawn correctly (steepest or shallowest possible line). All error bars must be plotted.	1
2(c)(iii)	Gradient determined with a triangle that is at least half the length of the drawn line. Must be negative.	1
	uncertainty = gradient of line of best fit – gradient of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line gradient – shallowest worst line gradient)	1
2(c)(iv)	y-intercept determined from substitution into $y = mx + c$.	1
	y-intercept determined using gradient of worst acceptable line. uncertainty = y-intercept of line of best fit – y-intercept of worst acceptable line or uncertainty = $\frac{1}{2}$ (steepest worst line y-intercept – shallowest worst line y-intercept) No ECF from false origin method.	1

Question	Answer	Marks
2(d)(i)	m determined using candidate's y -intercept and correct units for m and s . $m = \frac{y\text{-intercept}}{g} = \frac{y\text{-intercept}}{9.81}$	1
	s determined using candidate's gradient and m <u>and</u> s given to 2 or 3 significant figures. Correct substitution of numbers must be seen. $s = \frac{-2mg}{\text{gradient}} = \frac{-2 \times y\text{-intercept}}{\text{gradient}}$	1
2(d)(ii)	percentage uncertainty in m = percentage uncertainty in y -intercept	1
	percentage uncertainty in s = percentage uncertainty in gradient + percentage uncertainty in y -intercept or percentage uncertainty in s = percentage uncertainty in gradient + percentage uncertainty m Maximum/minimum methods: $\text{max } s = \frac{-2 \times \text{max } y\text{-intercept}}{\text{min gradient}} \text{ or } \frac{-2g \times \text{max } m}{\text{min gradient}}$ $\text{min } s = \frac{-2 \times \text{min } y\text{-intercept}}{\text{max gradient}} \text{ or } \frac{-2g \times \text{min } m}{\text{max gradient}}$ Correct substitution of numbers must be seen.	1